AI Intelligence Data Pipeline

Production Architecture for the AI Engineer Demo Task

Scale: Async workers behind a queue can scale from the demo to 500,000+ records without changing crawler logic. PostgreSQL stores canonical entities and provenance; Redis handles distributed deduplication, locks and rate-limit state; object storage retains raw snapshots.

Ingestion: Prefer official APIs and RSS/Atom feeds. Use bounded asyncio concurrency for permitted HTTP crawling. JavaScript-rendered sources can use Playwright where automation is allowed.

Freshness: News and jobs pass only when publication time is known and falls within the last 24 hours. Date extraction uses JSON-LD, metadata, RSS/Atom, visible dates and relative-date parsing. URL/title hashes prevent duplicate processing.

LLM: Gemini → Groq → DeepSeek. Clean HTML is chunked with a hard size budget. 429 responses use Retry-After or exponential backoff plus jitter. Oversized payloads are reduced and retried through the fallback chain.

Entity resolution: Normalize legal suffixes and punctuation, match aliases, then use conservative fuzzy matching. Every raw-to-canonical mapping records its method and score.

Anti-bot: The design does not attempt CAPTCHA solving or security-control bypass. Protected sources should be accessed through approved APIs, licensed feeds, or permitted browser automation.

Observability: Structured logs capture job ID, source, URL, status, retry count, provider, latency and validation errors. Metrics include 429/413 rates, freshness pass rate, duplicate rate and fallback rate.